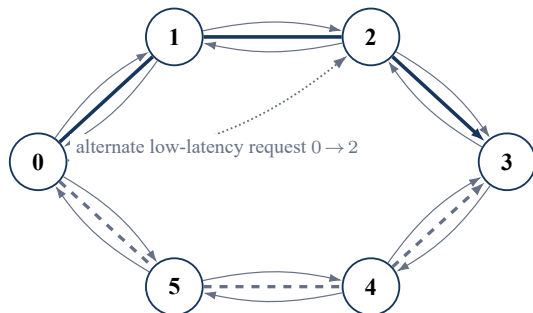


(a) Directed Ring-6 topology and four-slot candidate contract



Normal request: $0 \rightarrow 3$

Slot 0: clockwise candidate

$0 \rightarrow 1 \rightarrow 2 \rightarrow 3$ (solid)

Slot 1: counter-clockwise candidate

$0 \rightarrow 5 \rightarrow 4 \rightarrow 3$ (dashed)

Slots 2 and 3: reserved / invalid

Four-slot hardware interface remains fixed

Thin paired arrows show the 12 directed ring edges.
The emphasized routes are simple-path candidates,
not quantum repeaters.

(b) Repeating 32-decision trace phases

Nominal
32 decisions

Scarcity
32 decisions

Route-quality
stress
32 decisions

High load
32 decisions

Low latency
32 decisions

Mixed
32 decisions

Episode length: 512 decisions (16 phases); the six-phase schedule repeats.

Observe state
and decide

On success, consume
one key unit per se-
lected route link

Apply integer key arrivals
and registered link updates

Update state for
next decision